CS 210: Worksheet 19 *Dr. Mudassir Shabbir, October 29, 2026*

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Sequences, summations, and series. Adapted from Rosen, *Discrete Mathematics and Its Applications*, 7th ed., § 2.4. Show your working in the spaces provided.

1. Terms of a sequence *Easy*

For each sequence,[1] find a_0, a_1, a_2, a_3.

(a) $a_n = (n+1)^{n+1}$.

(b) $a_n = \lfloor n/2 \rfloor$.

(c) $a_n = 2^n + (-2)^n$.

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2. Finding a sequence rule *Moderate*

For each list, propose a simple explicit formula[2] for a_n, with the first term indexed by $n = 1$. Check your formula against the displayed terms, then use it to find the next three terms.

(a) $3, 5, 8, 12, 17, 23, 30, 38, 47, \ldots$

(b) $2, 16, 54, 128, 250, 432, 686, \ldots$

[1] A *sequence* is a function whose domain is a set of integer indices, usually $\{0, 1, 2, \ldots\}$ or $\{1, 2, 3, \ldots\}$. The value a_n is its term at index n. The floor $\lfloor x \rfloor$ is the greatest integer less than or equal to x.

[2] An *explicit formula* gives a_n directly in terms of n. A finite list can fit different rules, so your continuation depends on the rule you choose.

3. Sequences from recurrence relations *Easy*

Find the first five terms, starting at a_0, of each sequence defined by a recurrence relation and initial values.[3]

(a) $a_0 = 2$, and $a_n = 6a_{n-1}$ for $n \geq 1$.

(b) $a_0 = 1$, $a_1 = 2$, and $a_n = a_{n-1} + 3a_{n-2}$ for $n \geq 2$.

4. Summing over an index set *Easy*

Let $S = \{1, 3, 5, 7\}$. Expand each sum[4] and evaluate it. Show which terms are being added.

(a) $\displaystyle\sum_{j \in S} j$.

(b) $\displaystyle\sum_{j \in S} j^2$.

(c) $\displaystyle\sum_{j \in S} \frac{1}{j}$.

[3] A *recurrence relation* expresses a term of a sequence using earlier terms. *Initial values* supply the starting terms needed to apply the rule.

[4] The notation $\sum_{j \in S} f(j)$ adds one value $f(j)$ for each element j of the finite set S. The index visits only members of S.

5. Combining geometric sums

Moderate

Evaluate each sum by separating it into geometric sums.[5] Show the two sums you use before calculating the result.

(a) $\sum_{j=0}^{8}(3^j - 2^j)$.

(b) $\sum_{j=0}^{8}(2 \cdot 3^j + 3 \cdot 2^j)$.

6. Double summations

Moderate

Evaluate each double sum.[6] Show the inner sum for each value of the outer index.

(a) $\sum_{i=1}^{3}\sum_{j=1}^{2}(i - j)$.

(b) $\sum_{i=0}^{2}\sum_{j=0}^{3} i^2 j^3$.

[5] A *finite series* is a sum of finitely many sequence terms. It is *geometric* when successive terms have a fixed common ratio.

[6] In a *double sum*, evaluate the inner sum while holding the outer index fixed, then add those results over the outer index.

7. Cancellation two indices apart *Moderate*

Use cancellation to simplify the following sums.[7] Assume $n \geq 1$.

(a) Find a constant C such that $\dfrac{1}{k(k+2)} = C\left(\dfrac{1}{k} - \dfrac{1}{k+2}\right)$ for every positive integer k.

(b) Use part (a) to find a formula for $\displaystyle\sum_{k=1}^{n} \frac{1}{k(k+2)}$. Show which terms cancel and which remain. Check your formula when $n = 1$.

8. Infinite geometric series *Challenge*

For each infinite series,[8] find the partial sum S_n through index n. Decide whether the partial sums approach a finite number and, if so, find that number. You may use the fact that r^{n+1} approaches 0 when $|r| < 1$.

(a) $\displaystyle\sum_{k=2}^{\infty} 3\left(\frac{1}{4}\right)^k$.

(b) $\displaystyle\sum_{k=1}^{\infty} \left(-\frac{1}{3}\right)^k$.

(c) $\displaystyle\sum_{k=0}^{\infty} (-1)^k$.

[7] A *telescoping sum* has terms that cancel upon expansion, leaving only boundary terms.

[8] For $\sum_{k=m}^{\infty} a_k$, the *partial sum* is $S_n = \sum_{k=m}^{n} a_k$. The infinite series *converges* to L if S_n approaches the finite number L as n grows without bound; otherwise it *diverges*. The value of a convergent series is L.